

Introduction to Statistics - Homework #11

Exercise 8.33* - Husbands and wives, Part II

Data on **12 married couples** in Britain consist of the husband's height (`husband`, in cm) and the wife's height (`wife`, in cm). The wife's height is treated as the response variable, y_i , and the husband's height as the predictor variable, x_i . We consider the simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

Answer the following questions based on the following R output.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  80.5200    16.8278   4.785 0.000740 ***
husband       0.4805     0.0941   5.109 0.000458 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 2.324 on 10 degrees of freedom
Multiple R-squared:  0.7230, Adjusted R-squared:  0.6953
F-statistic: 26.10 on 1 and 10 DF, p-value: 0.0004583
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(1) Fill in the values shown in the table below.

$\hat{\beta}_0$	$\hat{\beta}_1$	Estimated Regression Equation	$MSE = \hat{\sigma}^2$

(2) A married couple is selected in which the husband is $x_i = 185$ cm tall, and the wife's actual height is $y_i = 173$ cm. Use the model to predict the mean wife's height $\hat{y}_i$ at $x_i = 185$ and compute the residual e_i .

(3) Using the R output from part (1), test whether the husband's height (`husband`, x_i) is a statistically significant predictor variable for the wife's height (`wife`, y_i) at the significance level $\alpha = 0.01$. Answer the following.

(a) State the null and alternative hypotheses.

(b) Report the p-value, and complete the hypothesis test. State the conclusion in the context of the data.

Exercise 8.23 - The Coast Starlight, Part II

The Coast Starlight Amtrak train runs from Seattle to Los Angeles. Using a linear model, we try to predict the amount of time it takes to travel from one stop to another (y , in minutes) using the distance between each stop (x , in miles). The mean travel time from one stop to the next on the Coast Starlight is $\bar{y} = 129$ minutes, with a standard deviation of $s_y = 113$ minutes. The mean distance traveled from one stop to the next is $\bar{x} = 108$ miles with a standard deviation of $s_x = 99$ miles. The correlation between travel time and distance is $r = 0.636$.

(1) Compute $\hat{\beta}_0$ and $\hat{\beta}_1$. Write the regression equation using $\hat{\beta}_0$ and $\hat{\beta}_1$.

(2) Calculate the coefficient of determination R^2 . Interpret R^2 in the context of the application.

(3) The distance between Santa Barbara and Los Angeles is $x_i = 103$ miles. Use the model to estimate the average time $\hat{y}_i$ it takes to travel between these two cities.

(4) It actually takes the Coast Starlight about $y_i = 168$ minutes to travel from Santa Barbara to Los Angeles. Calculate the residual e_i .

(5) Suppose Amtrak is considering adding a stop to the Coast Starlight $x = 500$ miles away from Los Angeles. Would it be appropriate to use this linear model to predict the travel time from Los Angeles to this point?